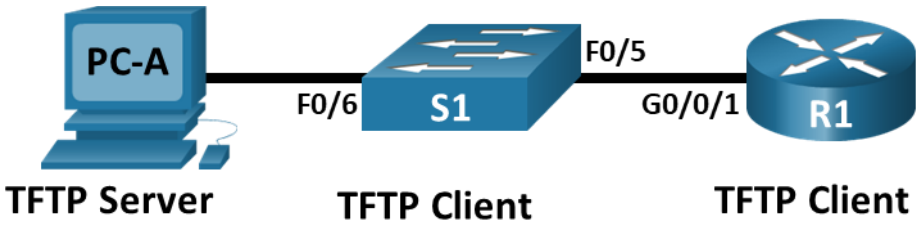
Lab - Use TFTP and Flash to Manage Configuration Files

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0/1	192.168.1.1	255.255.255.0	N/A
S1	VLAN 1	192.168.1.11	255.255.255.0	192.168.1.1
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1

Objectives

Part 1: Build the Network and Configure Basic Device Settings

Part 2: Use TFTP to Back Up and Restore the Switch Running Configuration

Part 3 Use TFTP to Back Up and Restore the Router Running Configuration

Part 4: Back Up and Restore Running Configurations Using Router Flash Memory

Background / Scenario

Cisco networking devices are often upgraded or swapped out for a number of reasons. It is important to maintain backups of the latest device configurations, as well as a history of configuration changes. A TFTP server is often used to backup configuration files and IOS images in production networks. A TFTP server is a centralized and secure method used to store the backup copies of the files and restore them as necessary. Using a centralized TFTP server, you can back up files from many different Cisco devices.

In this lab, you will use TFTP server software to back up the Cisco device running configuration to the TFTP server. You can edit the file using a text editor and copy the new configuration back to a Cisco device. The instructions here for the TFTP server configuration and operation are generic and there may be some differences in terminology with your TFTP server software.

Note: The routers used with CCNA hands-on labs are Cisco 4221 with Cisco IOS XE Release 16.9.4 (universalk9 image). The switches used in the labs are Cisco Catalyst 2960s with Cisco IOS Release 15.2(2) (lanbasek9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

Note: Ensure that the routers and switches have been erased and have no startup configurations. If you are unsure contact your instructor.

© 2013 - 2022 Cisco and/or its affiliates. All rights reserved. Cisco Public

Required Resources

- 1 Router (Cisco 4221 with Cisco IOS XE Release 16.9.3 universal image or comparable)
- 1 Switch (Cisco 2960 with Cisco IOS Release 15.2(2) lanbasek9 image or comparable)
- 1 PC (Windows with a terminal emulation program, such as Tera Term)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Instructions

Part 1: Build the Network and Configure Basic Device Settings

In Part 1, you will set up the network topology and configure basic settings, such as the interface IP addresses for router R1, switch S1 and PC-A.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram, and cable as necessary.

Step 2: Configure basic settings for the router.

a. Assign a device name to the router.

b. Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.

c. Assign **class** as the privileged EXEC encrypted password.

d. Assign **cisco** as the console password and enable login.

e. Assign **cisco** as the VTY password and enable login.

f. Encrypt the plaintext passwords.

g. Create a banner that warns anyone accessing the device that unauthorized access is prohibited.

h. Configure interfaces as listed in the table above.

i. Save the running configuration to the startup configuration file.

Note: Use the question mark (**?**) to help with the correct sequence of parameters needed to execute this command.

Step 3: Configure basic settings for the switch.

a. Assign a device name to the switch.

b. Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.

c. Assign **class** as the privileged EXEC encrypted password.

d. Assign **cisco** as the console password and enable login.

e. Assign **cisco** as the VTY password and enable login.

f. Encrypt the plaintext passwords.

g. Create a banner that warns anyone accessing the device that unauthorized access is prohibited.

h. Shut down all unused interfaces.

© 2013 - 2022 Cisco and/or its affiliates. All rights reserved. Cisco Public

i. Configure interface VLAN 1 as specified in the table above.

j. Save the running configuration to the startup configuration file.

Note: Use the question mark (**?**) to help with the correct sequence of parameters needed to execute this command.

Step 4: Verify connectivity from PC-A.

a. Ping from PC-A to S1.

b. Ping from PC-A to R1.

If the pings are not successful, troubleshoot the basic device configurations before continuing.

Part 2: Use TFTP to Back Up and Restore the Switch Running Configuration

Step 1: Verify connectivity to switch S1 from PC-A.

The TFTP application uses the UDP Layer 4 transport protocol, which is encapsulated in an IP packet. For TFTP file transfers to function, there must be Layer 1 and 2 (Ethernet, in this case) and Layer 3 (IP) connectivity between the TFTP client and the TFTP server. The LAN topology in this lab uses only Ethernet at Layers 1 and 2. However, TFTP transfers can also be accomplished over WAN links that use other Layer 1 physical links and Layer 2 protocols. As long as there is IP connectivity between the client and server, as demonstrated by ping, the TFTP transfer can take place. If the pings are not successful, troubleshoot the basic device configurations before continuing.

Note: A common misconception is that you can TFTP a file over the console connection. This is not the case because the console connection does not use IP. The TFTP transfer can be initiated from the client device (router or switch) using the console connection, but there must be IP connectivity between the client and server for the file transfer to take place.

Step 2: Start the TFTP server.

Start the TFTP program on PC-A. Ensure that the TFTP program is using a directory that you have WRITE permission for, such as a folder on your desktop.

Om tftpd64 inte finns på datorn laddas det ned via internet.

Step 3: Explore the copy command on a Cisco device.

a. Console into switch S1 and, from the privileged EXEC mode prompt, enter **copy ?** to display the options for source or “from” location and other available copy options. You can specify **flash:** or **flash0:** as the source. However, if you simply provide a filename as the source, **flash0:** is assumed and is the default. Note that **running-config** is also an option for the source location.

```
S1# copy ?
  /erase          Erase destination file system.
  /error          Allow to copy error file.
  /noverify       Don't verify image signature before reload.
  /verify         Verify image signature before reload.
  bs:             Copy from bs: file system
  cns:            Copy from cns: file system
  flash:          Copy from flash: file system
  ftp:            Copy from ftp: file system
  http:           Copy from http: file system
  https:          Copy from https: file system
  logging         Copy logging messages
```

© 2013 - 2022 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
  null:           Copy from null: file system
  nvram:          Copy from nvram: file system
  rcp:            Copy from rcp: file system
  running-config  Copy from current system configuration
  scp:            Copy from scp: file system
  startup-config  Copy from startup configuration
  system:         Copy from system: file system
  tar:            Copy from tar: file system
  tftp:           Copy from tftp: file system
  tmpsys:         Copy from tmpsys: file system
  vb:             Copy from vb: file system
  xmodem:         Copy from xmodem: file system
  ymodem:         Copy from ymodem: file system
```

b. Use the **?** to display the destination options after a source file location is chosen. The **flash:** file system for S1 is the source file system in this example.

```
S1# copy flash: ?
  flash:          Copy to flash: file system
  ftp:            Copy to ftp: file system
  http:           Copy to http: file system
  https:          Copy to https: file system
  null:           Copy to null: file system
  nvram:          Copy to nvram: file system
  rcp:            Copy to rcp: file system
  running-config  Update (merge with) current system configuration
  scp:            Copy to scp: file system
  startup-config  Copy to startup configuration
  system:         Copy to system: file system
  tftp:           Copy to tftp: file system
  tmpsys:         Copy to tmpsys: file system
  vb:             Copy to vb: file systesystem
```

Step 4: Transfer the running-config file from switch S1 to TFTP server on PC-A.

a. From the privileged EXEC mode on the switch, enter the **copy running-config tftp:** command. Provide the remote host address of the TFTP server (PC-A), 192.168.1.3. Press Enter to accept default destination filename (**s1-confg**) or provide your own filename. The exclamation marks (**!!**) indicate the transfer process is in progress and is successful.

```
S1# copy running-config tftp:
Address or name of remote host []? 192.168.1.3
Destination filename [s1-confg]?
!!
1465 bytes copied in 0.663 secs (2210 bytes/sec)
S1#
```

The TFTP server may also displays transfer progress.

Note: If you do not have permission to write to the current directory that is used by the TFTP server, the following error message displays:

```
S1# copy running-config tftp:
Address or name of remote host []? 192.168.1.3
```

© 2013 - 2022 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
Destination filename [s1-confg]?
%Error opening tftp://192.168.1.3/s1-confg (Permission denied)
```

Note: Other issues, such as a firewall blocking TFTP traffic, can prevent the TFTP transfer. Please check with your instructor for further assistance.

b. Check the directory on the TFTP server (usually the default directory for the TFTP server software) to verify that the file was transferred successfully. Your TFTP server may have a dialog for this, or you can simply use the File Explorer provided by your Operating System.

Step 5: Create a modified switch running configuration file.

The saved running configuration file, **s1-confg**, can also be restored to the switch by using the **copy** command from the switch. The original or a modified version of the file can be copied to the flash file system of the switch.

a. Navigate to the TFTP directory on PC-A by using the file system of PC-A, and then locate the **s1-confg** file. Open this file using a text editor program, such as WordPad.

b. With the file open, locate the **hostname S1** line. Replace **S1** with **Switch1**. Delete all the self-generated crypto keys, as necessary. A sample of the keys is displayed below. These keys are not exportable and can cause errors while updating the running configuration.

```
crypto pki trustpoint TP-self-signed-1566151040
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-1566151040
 revocation-check none
 rsakeypair TP-self-signed-1566151040
!
!
crypto pki certificate chain TP-self-signed-1566151040
 certificate self-signed 01
  3082022B 30820194 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
  31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
<output omitted>
  E99574A6 D945014F B6FE22F3 642EE29A 767EABF7 403930CA D2C59E23 102EC12E
  02F9C933 B3296D9E 095EBDAF 343D17F6 AF2831C7 6DA6DFE3 35B38D90 E6F07CD4
  40D96970 A0D12080 07A1C169 30B9D889 A6E2189C 75B988B9 0AF27EDC 6D6FA0E5
  CCFA6B29 729C1E0B 9DADACD0 3D7381
       quit
```

c. Save this file as a plain text file with a new filename, **Switch1-confg.txt**, in this example.

Note: When saving the file, an extension, such as **.txt**, may be added to the filename automatically.

d. If your TFTP software has the option, use it to show the contents of its directory to verify that the file is present.

Step 6: Copy the modified running configuration file from the TFTP server to switch S1.

a. From the privileged EXEC mode on the switch, enter the **copy tftp running-config** command. Provide the remote host address of the TFTP server, 192.168.1.3. Enter the new filename, **Switch1-confg.txt**. The exclamation mark (!) indicates the transfer process is in progress and is successful.

```
S1# copy tftp: running-config
Address or name of remote host []? 192.168.1.3
Source filename []? Switch1-confg.txt
Destination filename [running-config]?
```

© 2013 - 2022 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
Accessing tftp://192.168.1.3/Switch1-confg.txt...
Loading Switch1-confg.txt from 192.168.1.3 (via Vlan1): !
[OK - 1580 bytes]
[OK]
1580 bytes copied in 9.118 secs (173 bytes/sec)
*Mar  1 00:21:16.242: %PKI-4-NOAUTOSAVE: Configuration was modified.  Issue "write
memory" to save new certificate
*Mar  1 00:21:16.251: %SYS-5-CONFIG_I: Configured from tftp://192.168.1.3/Switch1-
confg.txt by console
Switch1#
```

After the transfer has completed, the prompt has changed from S1 to Switch1, because the running configuration is updated with the **hostname Switch1** command in the modified running configuration.

b. Enter the **show running-config** command to examine running configuration file.

```
Switch1# show running-config
Building configuration...

Current configuration : 3062 bytes
!
! Last configuration change at 00:09:34 UTC Mon Mar 1 1993
!
version 15.0
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname Switch1
!
boot-start-marker
boot-end-marker
<output omitted>
```

Note: This procedure merges the running-config from the TFTP server with the current running-config in the switch or router. If changes were made to the current running-config, the commands in the TFTP copy are added. Alternatively, if the same command is issued, it updates the corresponding command in the switch or router current running-config.

If you want to completely replace the current running-config with the one from the TFTP server, you must erase the switch startup-config and reload the device. You will then need to configure the VLAN 1 management address, so there is IP connectivity between the TFTP server and the switch.

Part 3: Use TFTP to Back Up and Restore the Router Running Configuration

The backup and restore procedure from Part 3 can also be performed with a router. In Part 4, the running configuration file will be backed up and restored using a TFTP server.

Step 1: Verify connectivity to router R1 from PC-A.

If the pings are not successful, troubleshoot the basic device configurations before continuing.

Step 2: Transfer the running configuration from router R1 to TFTP server on PC-A.

© 2013 - 2022 Cisco and/or its affiliates. All rights reserved. Cisco Public

a. From the privileged EXEC mode on R1, enter the **copy running-config tftp** command. Provide the remote host address of the TFTP server, 192.168.1.3, and accept the default filename.

b. Verify that the file has been transferred to the TFTP server.

Step 3: Restore the running configuration file to the router.

a. Erase the startup-config file on the router.

b. Reload the router.

c. Configure the G0/0/1 interface on the router with an IP address 192.168.1.1.

d. Verify connectivity between the router and PC-A.

e. Use the **copy** command to transfer the running-config file from the TFTP server to the router. Use **running-config** as the destination.

f. Verify the router has updated the running-config.

Part 4: Back Up and Restore Configurations Using Router Flash Memory

Current generation Cisco routers do not have internal flash memory. The flash memory for these routers uses CompactFlash (CF) memory. The use of CF memory allows for more available flash memory and easier upgrades without the need to open the router case. Besides storing the necessary files, such as IOS images, the CF memory can store other files, such as a copy of the running configuration. In Part 5, you will create a backup copy of the running configuration file and save it on the USB memory on the router.

Note: If the router does not use CF, the router may not have enough flash memory for storing the backup copy of running configuration file. You should still read through the instructions and become familiar with the commands.

Step 1: Display the router file systems.

The **show file systems** command displays the available file systems on the router. The **flash0:** file system is the default file system on this router as indicated by the asterisk (*) symbol (at the beginning of the line). The **flash0:** file system can also be referenced using the name **flash:**. The total size of the **flash0:** is approximately 7GB with about 6GB available. Currently **flash0:** and **nvram:** are the only available file systems.

```
R1# show file systems
File Systems:

       Size(b)       Free(b)      Type  Flags  Prefixes
             -             -    opaque     rw   system:
             -             -    opaque     rw   tmpsys:
*   7194652672    6299918336      disk     rw   bootflash: flash:
    1804468224    1723789312      disk     ro   webui:
             -             -    opaque     rw   null:
             -             -    opaque     ro   tar:
             -             -   network     rw   tftp:
             -             -    opaque     wo   syslog:
      33554432      33543116     nvram     rw   nvram:
             -             -   network     rw   rcp:
             -             -   network     rw   ftp:
             -             -   network     rw   http:
             -             -   network     rw   scp:
             -             -   network     rw   sftp:
```

© 2013 - 2022 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
-            -   network     rw   https:
-            -    opaque     ro   cns:
```

Where is the startup-config file located?

Note: Verify there is at least 1 MB (1,048,576 bytes) of free space. If there is not enough space in the flash memory, please contact your instructor for further instructions. You can determine the size of flash memory and space available using the **show flash** or **dir flash:** command at the privileged EXEC prompt.

Step 2: Copy the router running configuration to flash.

A file can be copied to flash by using the **copy** command at the privileged EXEC prompt. In this example, the file is copied into **flash0:**, because there is only one flash drive available as displayed in the previous step, and it is also the default file system. The **R1-running-config-backup** file is used as the filename for the backup running configuration file.

Note: Remember that filenames are case-sensitive in the IOS file system.

a. Copy the running configuration to flash memory.

```
R1# copy running-config flash:
Destination filename [running-config]? R1-running-config-backup
2169 bytes copied in 0.968 secs (2241 bytes/sec)
```

b. Use **dir** command to verify the running-config has been copied to flash.

```
R1# dir flash:
Directory of bootflash:/

   11  drwx            16384   Aug 2 2019 04:15:13 +00:00  lost+found
370945  drwx             4096  Sep 25 2019 20:17:11 +00:00  .installer
338689  drwx             4096   Aug 2 2019 04:15:55 +00:00  .ssh
217729  drwx             4096   Aug 2 2019 04:17:59 +00:00  core
379009  drwx             4096  Sep 25 2019 20:19:13 +00:00  .prst_sync
80641  drwx             4096   Aug 2 2019 04:16:09 +00:00  .rollback_timer
161281  drwx             4096   Aug 2 2019 04:16:11 +00:00  gs_script
112897  drwx            77824  Sep 25 2019 20:23:03 +00:00  tracelogs
362881  drwx             4096  Aug 23 2019 17:19:54 +00:00  .dbpersist
298369  drwx             4096   Aug 2 2019 04:16:41 +00:00  virtual-instance
   12  -rw-               30  Sep 25 2019 20:19:13 +00:00  throughput_monitor_params
 8065  drwx             4096   Aug 2 2019 04:17:55 +00:00  onep
   13  -rw-               35  Sep 25 2019 20:20:19 +00:00  pnp-tech-time
249985  drwx             4096  Aug 20 2019 17:40:11 +00:00  Archives
   14  -rw-            64414  Sep 25 2019 20:20:28 +00:00  pnp-tech-discovery-summary
   15  -rw-             3509  Sep 25 2019 20:24:32 +00:00  R1-running-config-backup
   17  -rw-          5032908  Sep 19 2019 14:16:23 +00:00
isr4200_4300_rommon_1612_1r_SPA.pkg
   18  -rw-        517153193  Sep 21 2019 04:24:04 +00:00  isr4200-
universalk9_ias.16.09.04.SPA.bin

7194652672 bytes total (6299643904 bytes free)
```

© 2013 - 2022 Cisco and/or its affiliates. All rights reserved. Cisco Public

c. Use the **more** command to view the running-config file in flash memory. Examine the file output and scroll to the Interface section. Notice the **no shutdown** command is not included with the GigabitEthernet0/1. The interface is shut down when this file is used to update the running configuration on the router.

```
R1# more flash:R1-running-config-backup
<output omitted>
interface GigabitEthernet0/1
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
<output omitted>
```

Step 3: Erase the startup configuration and reload the router.

Step 4: Restore the running configuration from flash.

a. Verify the router has the default initial configuration.

b. Copy the saved running-config file from flash to update the running-config.

```
Router# copy flash:R1-running-config-backup running-config
```

c. Use the **show ip interface brief** command to view the status of the interfaces. The interface GigabitEthernet0/1 was not enabled when the running configuration was updated, because it is administratively down.

```
R1# show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES unset  administratively down down
GigabitEthernet0/0/1   192.168.1.1     YES TFTP   administratively down down
Serial0/1/0            unassigned      YES unset  administratively down down
Serial0/1/1            unassigned      YES unset  administratively down down
```

The interface can be enabled using the **no shutdown** command in the interface configuration mode on the router.

Another option is to add the **no shutdown** command for the GigabitEthernet0/0/1 interface to the saved file before updating the router running configuration file. This will be done in Part 5 using a saved file on a USB flash drive.

Note: Because the IP address was configured by using a file transfer, TFTP is listed under the Method heading in the **show ip interface brief** output.

© 2013 - 2022 Cisco and/or its affiliates. All rights reserved. Cisco Public